SOCIAL MEDIA ENGAGEMENT ANALYSIS

Data Analytics Internship Project Report

Tools: Microsoft Excel Online, Python (Google Colab), and Microsoft Power BI
Dataset: Synthetic sample of 1,200 social media posts from 2025

Dataset disclosure: The data is synthetic and does not represent verified activity from a real company or social media account.

1. Project Overview

Problem Statement

This project analyzes social media engagement to compare platform performance, identify popular content and themes, and examine patterns across posting schedules. The goal is to turn engagement data into clear findings and practical ideas for further testing.

Objectives

- Prepare and validate the dataset.
- Calculate key engagement metrics.
- Compare engagement across platforms, content types, and themes.
- Analyze patterns by posting day, hour, and month.
- Build visualizations and an interactive Power BI dashboard.

2. Dataset and Methodology

The synthetic dataset contains 1,200 posts dated within 2025. Fields include post date and time, platform, content type, theme, likes, comments, shares, clicks, reach, impressions, and followers.

Tools Used

Excel Online	Data checks, pivot tables, and summary analysis
Python (Google Colab)	Validation, calculations, grouped analysis, and charts
Power BI Desktop	Interactive dashboard and visual reporting

Data Preparation

Checks covered missing values, duplicate records, post identifier uniqueness, negative numeric values, and date/time validity. Derived fields were created for day of week, month, posting hour, and time period. Total engagement and engagement rate were calculated, and key summaries were compared between Excel and Python.

Key Metrics

Total Engagement	Likes + Comments + Shares + Clicks
Engagement Rate (%)	(Total Engagement / Reach) × 100
Average Engagement	Total engagement across posts / Number of posts
Average Engagement Rate	Average of the individual post engagement rates

3. Analysis and Key Findings

Overall Performance

Total Posts	1,200
Total Engagement	1,536,305
Average Engagement per Post	1,280.25
Average Engagement Rate	5.25%

Platform Comparison

Posts	705	495
Total Engagement	1,228,784	307,521
Average Engagement	1,742.96	621.25
Average Engagement Rate	6.04%	4.12%

Instagram recorded higher average engagement and average engagement rate than Facebook in this synthetic sample.

Content and Theme Performance

- All ten posts ranked highest by total engagement were Instagram Reels; this is a sample-specific pattern, not a guarantee about other accounts.
- Behind the Scenes recorded the highest average engagement among themes (approximately 1,399.53).
- Customer Stories had an average engagement rate of 5.66%, while Announcements had the lowest rate among the listed themes (4.28%).

Posting Patterns

- Sunday had the highest average engagement rate among weekdays (5.64%); Wednesday had the highest average engagement per post by weekday (approximately 1,395.27).
- 7 PM had the highest average engagement among observed posting hours (approximately 1,591.03).
- Evening posts averaged 1,358.67 engagement actions; November had the highest average engagement per post among months (approximately 1,431.47).

These are descriptive findings from synthetic data. They do not prove that a platform, content type, theme, or posting time causes higher engagement.

4. Power BI Dashboard

The Power BI report contains three pages that present the analysis and support interactive exploration.

Executive Overview	KPI cards, engagement by platform and content type, monthly trend, and platform/content/date slicers
Content Performance	Average engagement by content type, engagement rate and total engagement by theme, and top ten posts
Posting Strategy	Engagement by weekday, hour, month, and time period, plus post counts

Dashboard Evidence

Figure 1. Executive Overview Dashboard

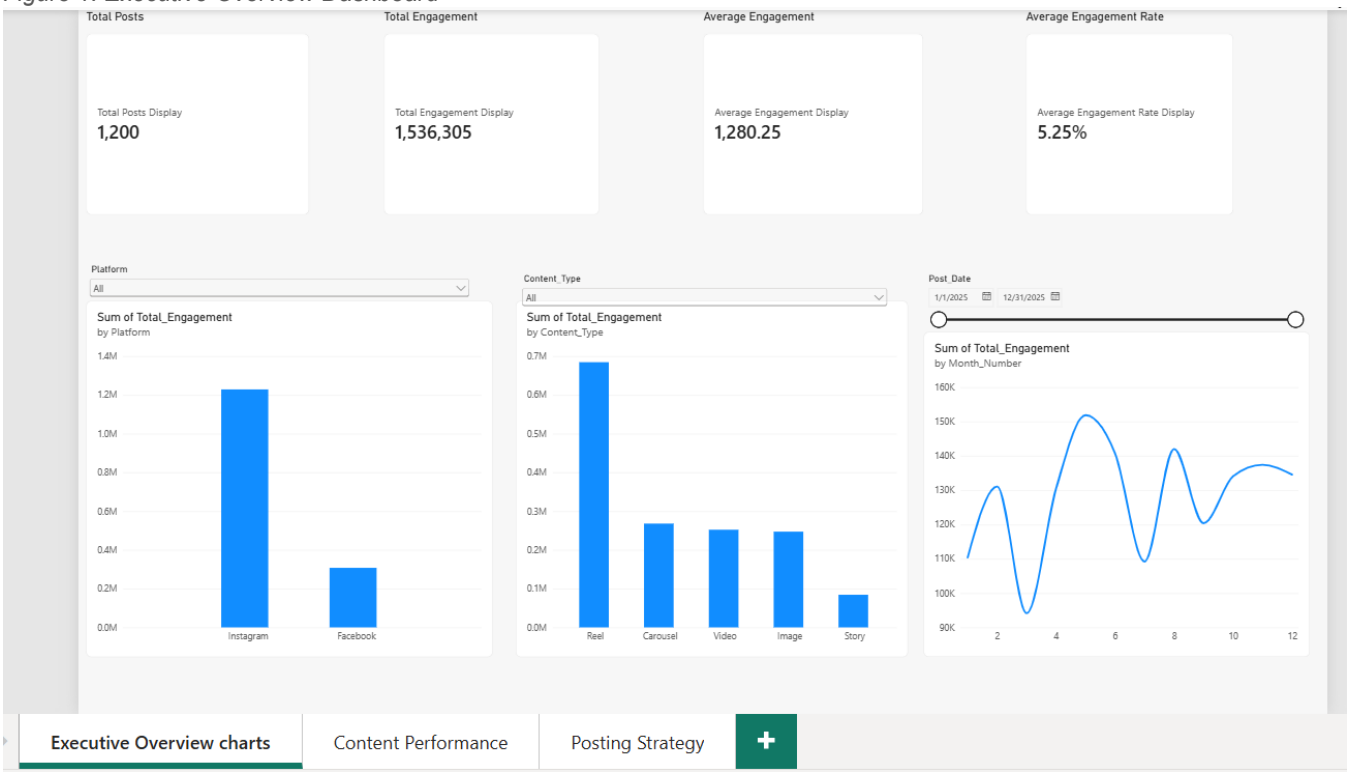


Figure 2. Content Performance Dashboard

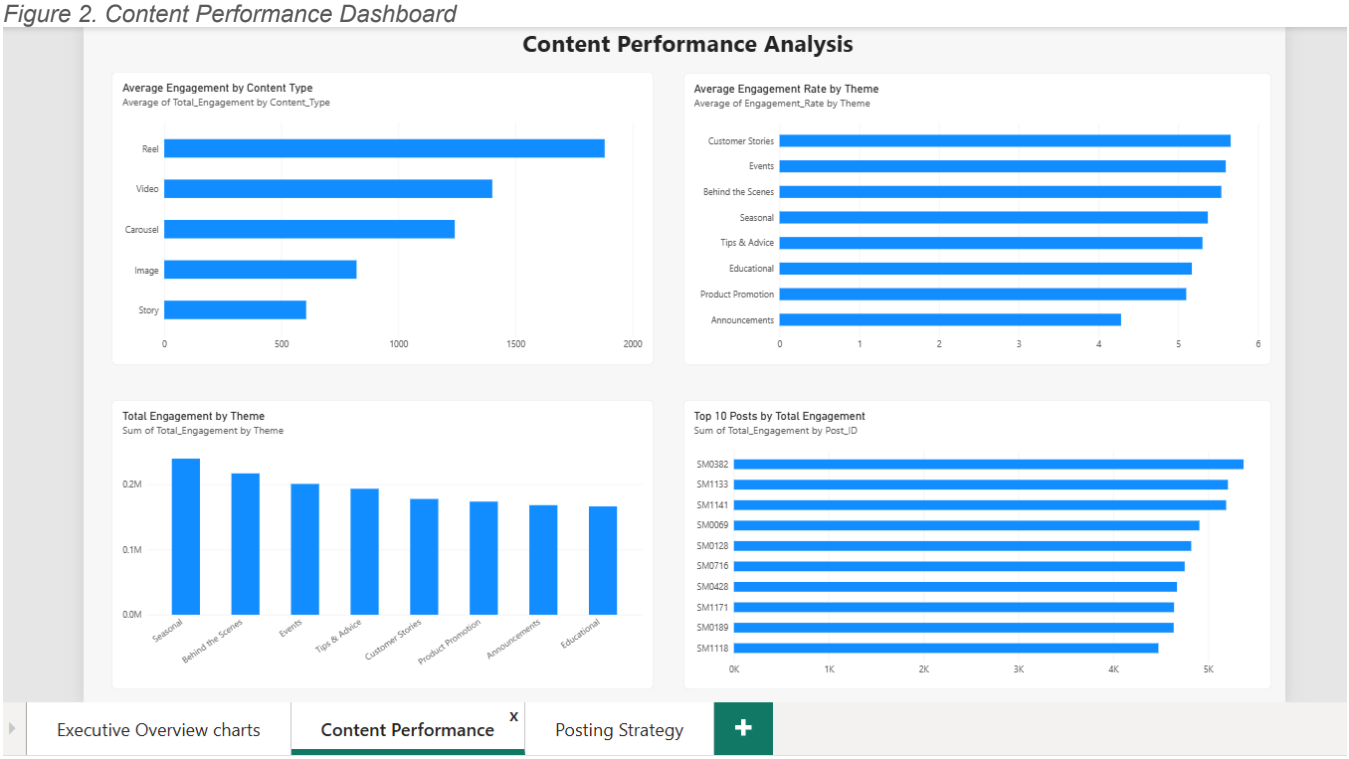
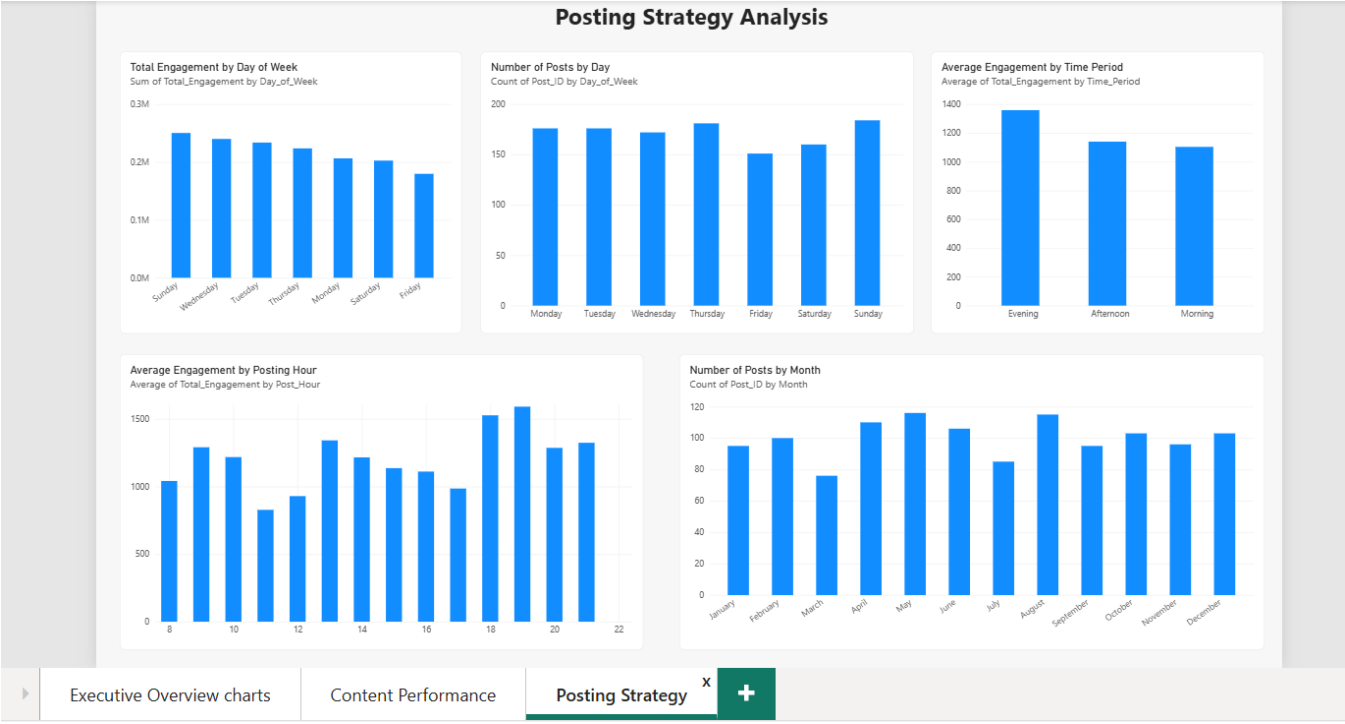


Figure 3. Posting Strategy Dashboard



5. Recommendations and Limitations

Recommendations

- Compare platform results using real account data before prioritizing one platform.
- Test Reels against other formats and compare engagement rate and reach over a representative period.
- Experiment with behind-the-scenes content and review the format and relevance of announcements.
- Test Sunday and 7 PM posting against alternative schedules; assess results over multiple weeks.
- Track engagement, engagement rate, and reach consistently using the same metric definitions.

These recommendations are hypotheses for testing, not guaranteed outcomes.

Limitations

- The dataset is synthetic and does not represent verified real-world social media activity.
- Posting volumes vary across categories and time periods, and some hours have no observations.
- The analysis is descriptive and cannot establish causal relationships.

6. Conclusion

This project demonstrates an end-to-end data analytics workflow, from data preparation and validation to exploratory analysis, visualization, and interactive dashboard development. The sample showed differences in engagement across platforms, themes, and posting times. Excel, Python, and Power BI were used to calculate metrics, identify patterns, and communicate findings. The results should be validated with real-world data before being used for business decisions.